

WiFi-Based 3D Human Pose Estimation Using CNN and Hybrid CNN-Transformer Architectures

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1. Introduction

In this paper, we explore WiFi-based human pose estimation (HPE) using the MM-Fi dataset [8] for 3D avatar simulation. Human bodies alter the propagation of WiFi signals in ways that are detectable through Channel State Information (CSI), and this encodes latent information about the pose of people in their environment. While deep learning methods such as HRNet [6] and OpenPose [1] have achieved strong performance on image-based pose estimation, they degrade under occlusion or poor lighting, conditions common in home and healthcare settings. WiFi-based approaches provide a compelling alternative: they are low-cost, leveraging existing infrastructure without additional sensing hardware. In addition, they avoid capturing personally identifying images. We will train a convolutional neural network and a hybrid CNN-transformer network to regress 3D human pose keypoints directly from WiFi CSI amplitude tensors, evaluating on MM-Fi’s synchronized WiFi and pose annotation data. We note that while WiFi-based sensing avoids camera-based privacy concerns, the covert nature of the sensing mechanism raises its own privacy considerations worth acknowledging.

2. Related work

Before WiFi-based pose estimation emerged as a research direction, a broader literature established that CSI signals encode meaningful information about human behavior through the task of activity recognition (HAR). Chen et al. demonstrated this with an attention-based bidirectional LSTM that classifies activities with 96.7% accuracy in an activity-room setting and 97.3% accuracy in a meeting-room setting [2], suggesting that CSI captures sufficient information about human movement to support more fine-grained inference tasks.

A significant body of work on WiFi-based HPE has emerged from Yang et al., whose MetaFi system [7] demonstrated that CSI signals can be mapped to 2D pose landmarks using WPNet, a convolutional network with residual blocks. Trained with cross-modal supervision from HRNet [6], MetaFi achieved a PCK@50 of 95.23%, establishing

a strong baseline for WiFi-based approaches. Their subsequent release of the MM-Fi dataset [8], which provides synchronized CSI and ground-truth pose annotations, enables direct comparison against this baseline without requiring cross-modal training. Yang et al. subsequently introduced MetaFi++ [9], augmenting the convolutional backbone with a self-attention transformer layer to achieve a PCK@50 of 97.30%, representing the current state of the art. Yang et al. later introduced MM-Fi [8], a large-scale multi-modal dataset for non-intrusive human pose estimation spanning four sensing modalities: RGB images, LiDAR point clouds, mmWave radar, and WiFi CSI. Benchmark experiments on MM-Fi demonstrated that fusing RGB, LiDAR, and WiFi CSI achieves higher 3D pose estimation accuracy than any single modality alone, highlighting the complementary nature of these signals.

Geng et al. [4] demonstrated an alternative formulation of WiFi-based pose estimation, mapping CSI phase and amplitude to UV coordinates, a dense representation of the correspondence between image pixels and a 3D human body surface mesh. Their architecture uses a two-branch encoder-decoder to translate CSI samples into 2D feature maps, which are then processed by a DensePose-RCNN to estimate per-body-part UV maps. While this approach produces richer output than keypoint-based methods, it requires phase sanitization preprocessing and a significantly more complex training pipeline.

Finally, Cho et al. [3] further motivate the real-world utility of WiFi-based pose estimation, proposing an architecture that combines convolutional encoders with Transformer-based attention mechanisms to estimate pose keypoints from CSI signals. The resulting estimates serve as input to a directed graph neural network (DGNN) for elderly fall detection, a concrete demonstration of the healthcare monitoring applications that motivate privacy-preserving, camera-free sensing approaches such as ours.

3. Method/Technical Approach

Our approach transforms raw WiFi Channel State Information (CSI) into 3D skeletal keypoints using a deep convolutional pipeline. Building upon the MetaFi system proposed

by Yang et al. [7], we extend the original architecture from 2D to 3D keypoint regression and additionally evaluate a second network architecture described in Section 3.2.

3.1. Dataset Utilization

The dataset we use is the MM-Fi dataset [8], which provides synchronized CSI packets and ground-truth pose annotations. For signal extraction we take the amplitude data from the CSI, as the phase shift is typically too noisy for reliable pose estimation. The raw CSI data is public from the experiment and structured into a $3 \times 114 \times T$ tensor, where 3 represents the receiver antennas, 114 represents the sub-carriers, and $T = 10$ represents the number of CSI packets synchronized to a single video frame. Finally, before entering the network, CSI features are upsampled, which helps prepare the data for the five successive CNN blocks to extract fine-grained patterns before it is downsampled in later stages. In addition, the MM-Fi dataset contains approximately 320,000 training instances across five sensing modalities. Since our approach uses only the WiFi CSI modality, the effective dataset size and computational requirements are significantly reduced. However, we still found that using high-performance NVIDIA A100 GPUs significantly improved training and inference speeds on the WiFi CSI data.

3.2. Network Architectures

We evaluate two distinct architectural paradigms to understand the trade-offs between local feature extraction and global context modeling.

3.2.1 Baseline CNN (WPNet)

Following the MetaFi implementation [7], our baseline processes WiFi CSI amplitude data through a single-branch convolutional network. The three receiver antenna streams are concatenated along the time axis, forming a $1 \times 114 \times 30$ tensor, which is then bilinearly upsampled to $1 \times 136 \times 136$. This passes through five convolutional blocks with channel depths increasing from 64 to 512: the first is a custom Conv+BN+ReLU block, while blocks 2–5 use pretrained ResNet-34 residual stages (3, 4, 6, 3 blocks respectively) with residual connections every two layers. A 1×1 convolutional bottleneck projects from 512 channels to 3 (x, y, z), followed by average pooling and a final transpose to regress 17×3 joint coordinates. No final ReLU is applied, as output coordinates can be negative in the root-centered coordinate frame.

3.2.2 MetaFi++ (WPFormer)

Building on WPNet, our extension implements the MetaFi++ architecture [9] [5], processing each antenna independently. Each of the three antenna streams is reshaped

to a $1 \times 136 \times 32$ tensor and passed through the same shared five-block ResNet-34 encoder. The resulting $512 \times 17 \times 4$ feature maps are concatenated along the width dimension to form a $512 \times 17 \times 12$ tensor, then normalized with BatchNorm. A ChannelTransformer block then processes these fused features: spatial locations are flattened into 204 tokens of dimension 512, learned positional embeddings are added, and a single encoder layer applies 3-head self-attention with InstanceNorm score normalization followed by a 2-layer FFN with GELU activation. Residual connections wrap both the attention and FFN sub-layers, and an additional outer residual skip connects the transformer input directly to its output. The attended features are reconstructed to spatial dimensions and passed through a convolutional decoder ($512 \rightarrow 32 \rightarrow 3$ channels, no final ReLU) and average pooling to regress 17×3 joint coordinates.

3.3. Loss Functions and Training

We train both WPNet and WPFormer by minimizing Mean Squared Error (MSE) between root-centered predicted coordinates $\hat{y}$ and root-centered ground truth annotations y from MM-Fi:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_i - y_i\|_2^2$$

where poses are centered at the root joint (pelvis, joint 0) prior to loss computation. This removes global position error from the training signal and focuses optimization on pose shape. We additionally experiment with a per-joint weighted variant that upweights arm joints. We upweight wrists and elbows (joints 12, 13, 15, 16) by a factor of 3 and shoulders (joints 11, 14) by a factor of 2 to address observed collapse in predicted variance at extremities, which we will discuss in later sections. Both models are optimized with SGD with momentum 0.9, batch size 32, and initial learning rate 0.001 for 50 epochs. WPNet uses MultiStepLR scheduling, halving the learning rate at epochs 10, 20, 30, and 40. WPFormer uses a linear decay schedule that holds the learning rate constant for the first 20 epochs then decays linearly to zero by epoch 50. Weight initialization is applied selectively to newly introduced layers only, preserving pretrained ResNet-34 weights in the encoder backbone. New convolutional layers use Xavier normal initialization, BatchNorm layers are initialized to the identity transformation ($\gamma = 1, \beta = 0$), and feed-forward layers in the transformer use Xavier uniform initialization with near-zero bias.

4. Evaluation Metric

To quantify the performance of our models for 3D human pose estimation, we utilize two standard metrics that assess accuracy at different levels of alignment: Mean Per-Joint

Position Error (MPJPE) and Procrustes-Aligned Mean Per-Joint Position Error (PA-MPJPE).

- **Mean Per-Joint Position Error (MPJPE):** This metric calculates the average Euclidean distance between the predicted 3D joint coordinates and the ground-truth annotations. To isolate the pose from absolute world coordinates, both the predicted and ground-truth skeletons are root-centered by subtracting the pelvis (root) joint position from all other joints before the error is computed.
- **Procrustes-Aligned MPJPE (PA-MPJPE):** In addition to root-alignment, this metric applies a Procrustes transformation which includes optimal rotation, translation, and scaling to the predicted pose to match the ground truth. This allows for an evaluation of the human pose structure independent of global orientation or positioning errors.

Both metrics are reported in millimeters (mm). Consistent with these evaluation metrics, both the baseline and hybrid models are trained using a root-centered Mean Squared Error (MSE) loss. This objective function encourages the network to prioritize learning the relative spatial structure of the 17 body keypoints over absolute room coordinates.

For generalization, we want to avoid a model that only performs well because it is implicitly memorizing subject-specific signal patterns. So, we will evaluate our models under a cross-subject split, which will test whether the model generalizes to unseen people rather than overfitting to the body proportions of the training subjects. Specifically, we use 32 subjects for training and 8 subjects for validation. We believe this results in more realistic benchmarking.

Qualitatively, we will present several types of figures. First, we will show a side-by-side visualization of predicted versus ground-truth skeletons on representative test frames. Second, we will show per-joint MPJPE and PA-MPJPE plots to visualize which body parts prove most challenging. Third, we will show training and validation curves to compare convergence behavior across architectures and losses. Finally, we will show case examples of failure, such as for fast limb motion, unusual body orientation, or environmental shifts when they occur.

5. Results

5.1. Baseline CNN Results

As shown in Figure 1 and Figure 2, we trained the baseline model for 20 epochs, stopping early after observing clear signs of convergence. During training, the training loss decreased steadily and consistently throughout all 20 epochs, dropping from an initial value of 0.0236 at epoch 1 to 0.0092 by epoch 20. However, the validation loss told a different story; it plateaued at approximately 0.0116-0.0120 after just 5 epochs of training, and remained essentially flat for the remainder of the training run. This divergence be-

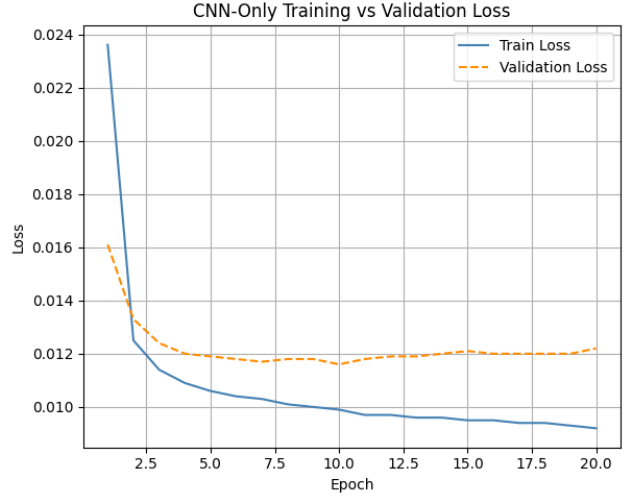


Figure 1. Baseline Training and Validation Loss

tween training and validation loss is characteristic of mild overfitting, where the model begins to memorize training-specific patterns that do not generalize to unseen subjects.

The application of learning rate decay at epoch 10, which reduced the learning rate from 0.001 to 0.0005, did not produce any meaningful improvement in validation performance as might be expected if the model were still actively learning. Instead, the validation loss remained flat and even increased slightly in subsequent epochs, further confirming that the model had reached its capacity limit on this task.

In terms of pose estimation metrics, the most meaningful improvements in MPJPE and PA-MPJPE were concentrated in the first 10 epochs of training, where MPJPE dropped substantially from 169.4mm at epoch 1 to 128.6mm at epoch 10, a reduction of over 40mm. After epoch 10, improvements became marginal, with MPJPE fluctuating within a narrow band of 128-131mm for the remainder of training. The best checkpoint was saved at epoch 10 with an MPJPE of 128.6mm and PA-MPJPE of 109.9mm, and this checkpoint was used for all subsequent evaluation and analysis.

5.2. Hybrid CNN-Transformer Results

As shown in Figure 3 and Figure 4, we trained the hybrid model for 30 epochs, stopping early after observing clear signs of convergence. During training, the training loss decreased steadily and consistently throughout all 30 epochs, dropping from an initial value of 0.0439 at epoch 1 to 0.0101 by epoch 30. However, the validation loss plateaued at approximately 0.0108-0.0110 after just 7 epochs of training, and remained essentially flat for the remainder of the training run. Again, this divergence between training and validation loss is characteristic of mild overfitting, where the model begins to memorize training-specific patterns that

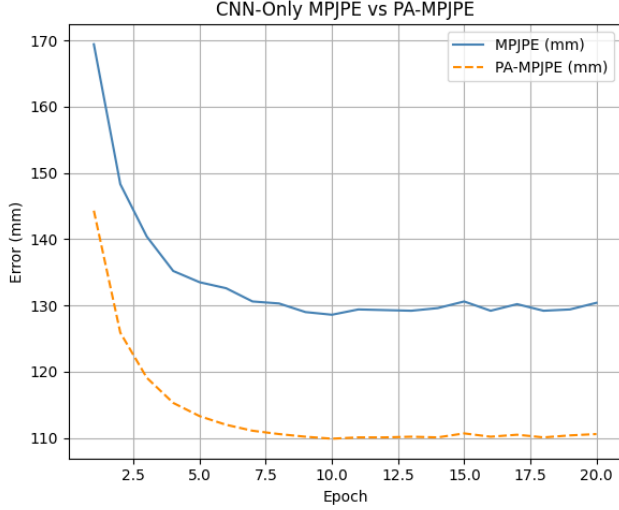


Figure 2. Baseline MPJPE and PA-MPJPE

do not generalize to unseen subjects. However, the overfitting is significantly less pronounced compared to the baseline training results, as evidenced by the smaller gap between training and validation loss.

Again, the application of linear learning rate decay at epoch 20, which reduced the learning rate did not produce any meaningful improvement in validation performance as might be expected if the model were still actively learning. Instead, the validation loss remained flat, further confirming that the model had reached its capacity limit on this task.

In terms of pose estimation metrics, the most meaningful improvements in MPJPE and PA-MPJPE were concentrated in the first 15 epochs of training, where MPJPE dropped substantially from 167.7mm at epoch 1 to 117.4mm at epoch 15, a reduction of over 50mm. After epoch 15, improvements became marginal, with MPJPE fluctuating within a narrow band of 116-118mm for the remainder of training. The best checkpoint was saved at epoch 18 with an MPJPE of 116.9mm and PA-MPJPE of 101.9mm, and this checkpoint was used for all subsequent evaluation and analysis.

5.3. Diagnosing Failure Modes

To better understand why our models struggle to generalize to unseen subjects, we visualized predicted skeleton keypoints projected onto the 2D X-Y plane. Representative visualizations for the hybrid CNN-Transformer architecture are shown in Figures 5 and 6. Across both models, we consistently observed that distal joints, particularly wrists and elbows, are poorly predicted. Rather than tracking the true joint positions, both models appear to regress these joints toward their mean position regardless of the input, indicating prediction collapse.

This behavior is a form of underfitting specific to indi-

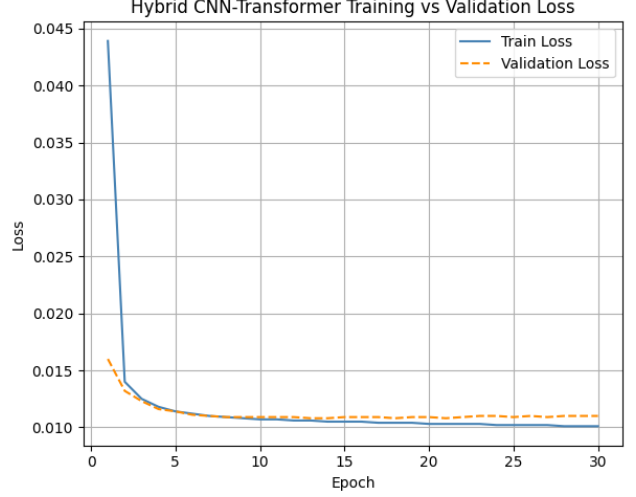


Figure 3. Hybrid CNN-Transformer Training and Validation Loss

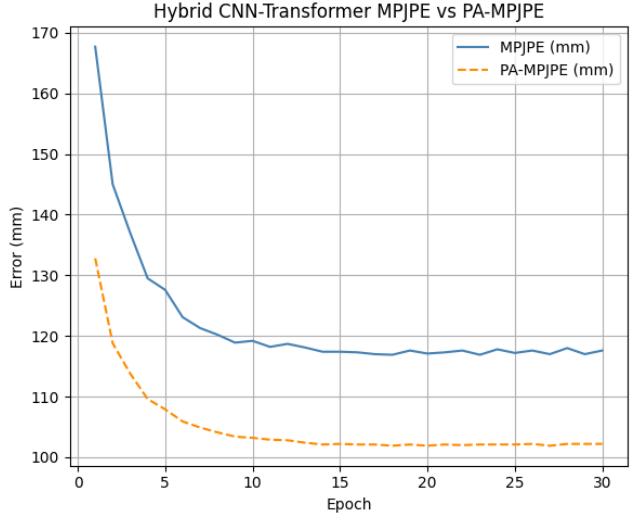


Figure 4. Hybrid CNN-Transformer MPJPE and PA-MPJPE

vidual joints: the model has failed to learn a discriminative mapping from CSI features to distal joint positions, and instead defaults to predicting a safe average. Such joint-level collapse typically arises from one or more of the following causes: insufficient training data for the affected joints, a lack of model expressiveness, or an absence of discriminative signal in the input modality itself. We investigate this behavior quantitatively in the following sections.

5.3.1 Per-Joint Standard Deviation

To quantify the observed prediction collapse, we computed the per-joint standard deviation of predictions across the validation set and compared it against the ground truth standard deviation for both models. As shown in Table 1, wrist

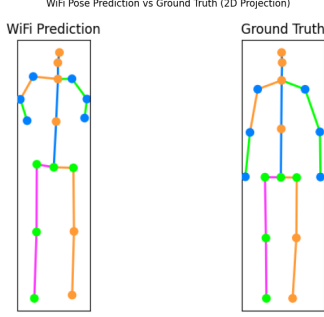


Figure 5. Hybrid CNN-Transformer Keypoint Visualization

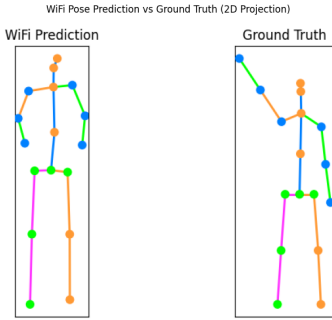


Figure 6. Hybrid CNN-Transformer Keypoint Visualization Depicting Failure Mode

and elbow joints exhibit a substantially lower predicted standard deviation than the corresponding ground truth, indicating that the model is producing near-constant predictions for these joints regardless of the input. For the baseline model, the left elbow has a predicted standard deviation of 0.0793 compared to a ground truth standard deviation of 0.1603, a ratio of 0.49x. The left wrist is even more collapsed, with a predicted standard deviation of 0.0729 against a ground truth of 0.2598, a ratio of just 0.28x. In contrast, joints corresponding to high-mass body segments show no such collapse. The spine has a predicted standard deviation of 0.1493 versus 0.1426 in the ground truth (1.05x), and the chest shows 0.2656 versus 0.2853 (0.93x). These joints are predicted with variance closely matching the ground truth distribution, demonstrating that the collapse is joint-specific rather than a global property of the model. As shown in Table 6, the hybrid CNN-Transformer model exhibited the same pattern, with wrist and elbow joints collapsing to similar ratios, suggesting the failure is not attributable to architectural choice but rather to the WiFi CSI modality.

Table 1. Baseline Per-Joint Standard Deviation: Prediction vs Ground Truth

Joint	Body Part	Pred Std	GT Std
0	Pelvis (root)	0.0000	0.0000
1	Left hip	0.0506	0.0509
2	Left knee	0.2433	0.2433
3	Left ankle	0.4294	0.4303
4	Right hip	0.0621	0.0509
5	Right knee	0.1276	0.1385
6	Right ankle	0.3092	0.3268
7	Spine	0.1493	0.1426
8	Chest/Thorax	0.2656	0.2853
9	Neck	0.3226	0.3463
10	Head	0.3454	0.3733
11	Right shoulder	0.3027	0.3224
12	Right elbow	0.2702	0.3239
13	Right wrist	0.2067	0.3416
14	Left shoulder	0.1839	0.1942
15	Left elbow	0.0793	0.1603
16	Left wrist	0.0729	0.2598

Table 2. Hybrid CNN-Transformer Per-Joint Standard Deviation: Prediction vs Ground Truth

Joint	Body Part	Pred Std	GT Std
0	Pelvis (root)	0.0000	0.0000
1	Left hip	0.0500	0.0509
2	Left knee	0.2470	0.2433
3	Left ankle	0.4390	0.4303
4	Right hip	0.0524	0.0509
5	Right knee	0.1374	0.1385
6	Right ankle	0.3244	0.3268
7	Spine	0.1450	0.1426
8	Chest/Thorax	0.2802	0.2853
9	Neck	0.3276	0.3463
10	Head	0.3612	0.3733
11	Right shoulder	0.3092	0.3224
12	Right elbow	0.2772	0.3239
13	Right wrist	0.2064	0.3416
14	Left shoulder	0.1836	0.1942
15	Left elbow	0.0751	0.1603
16	Left wrist	0.0609	0.2598

5.3.2 Per-Action MPJPE

To understand which actions were most affected by the collapse in distal joint predictions, we computed per-action MPJPE across the validation set for both models. As shown in Table 3, the baseline model performed best on Raising Hand (left) and Kicking (toward right), with MPJPE scores of 108.5mm and 108.9mm respectively. The hybrid CNN-Transformer model similarly performed best on Rais-

Table 3. Baseline Per-Action MPJPE

Action	Description	MPJPE (mm)
A02	Chest Expansion (horizontal)	129.7
A03	Chest expansion (vertical)	134.1
A04	Twist (left)	116.7
A05	Twist (right)	121.2
A13	Raising hand (left)	108.5
A14	Raising hand (right)	109.3
A17	Waving hand (left)	131.9
A18	Waving hand (right)	138.4
A19	Picking up things	189.6
A20	Throwing (toward left)	135.2
A21	Throwing (toward right)	132.4
A22	Kicking (toward left)	112.5
A23	Kicking (toward right)	108.9
A27	Bowing	131.4

ing Hand (left) and Raising Hand (right), achieving 92.2mm and 95.0mm as shown in Table 4. Across both models, Picking up Things was consistently the worst-performing action by a large margin, with MPJPE scores of 189.6mm and 194.8mm for the baseline and hybrid models respectively. These results are consistent with the joint-level analysis. Picking up Things requires simultaneous large displacements of both wrists and elbows, as well as a significant forward bend of the torso — a configuration that deviates substantially from the upright posture seen in most other actions. This compound difficulty likely explains the substantially higher error for this action. Raising Hand actions, despite involving wrist movement, are dominated by shoulder elevation which is a high-mass, large-amplitude motion that produces a strong CSI response. The model predicts the shoulder position well, keeping overall MPJPE low even if the wrist position itself is imprecise.

5.3.3 Loss Function Ablation

To determine whether our models possess the capacity to produce more accurate predictions for arm joints, we conducted an ablation in which wrist and elbow joints were upweighted by a factor of 3 and shoulder joints by a factor of 2 in the MSE loss function. We trained the hybrid CNN-Transformer model with the modified loss function for 10 epochs, stopping early upon observing that the validation loss had plateaued for several consecutive epochs. The checkpoint saved at epoch 10 achieved an MPJPE of 121.2mm and a PA-MPJPE of 104.0mm

As shown in Table 6, despite this ablation, the predicted standard deviation for wrist and elbow joints continued to collapse. This result suggests that the collapse is not a consequence of insufficient training signal for these joints, but rather reflects a fundamental limitation in the information

Table 4. Hybrid CNN-Transformer Per-Action MPJPE

Action	Description	MPJPE (mm)
A02	Chest Expansion (horizontal)	106.7
A03	Chest expansion (vertical)	115.9
A04	Twist (left)	102.9
A05	Twist (right)	108.8
A13	Raising hand (left)	92.2
A14	Raising hand (right)	95.0
A17	Waving hand (left)	117.7
A18	Waving hand (right)	121.5
A19	Picking up things	194.8
A20	Throwing (toward left)	124.6
A21	Throwing (toward right)	124.2
A22	Kicking (toward left)	100.0
A23	Kicking (toward right)	104.3
A27	Bowing	128.2

Table 5. Hybrid CNN-Transformer Per-Action MPJPE with Modified MSE Loss

Action	Description	MPJPE (mm)
A02	Chest Expansion (horizontal)	114.2
A03	Chest expansion (vertical)	123.4
A04	Twist (left)	109.9
A05	Twist (right)	112.2
A13	Raising hand (left)	96.4
A14	Raising hand (right)	102.8
A17	Waving hand (left)	118.1
A18	Waving hand (right)	129.4
A19	Picking up things	193.1
A20	Throwing (toward left)	128.1
A21	Throwing (toward right)	127.6
A22	Kicking (toward left)	103.7
A23	Kicking (toward right)	104.9
A27	Bowing	132.6

available to the model from the WiFi CSI input. In addition, Table 5 shows once again that Picking up Things was the worst-performing action by a large margin, with an MPJPE score of 193.1mm.

6. Conclusion

Our study demonstrates that WiFi Channel State Information can support rough 3D human pose estimation without the privacy concerns of traditional camera-based systems. We successfully implemented and evaluated two deep learning architectures, showing that the hybrid CNN-Transformer model outperformed the baseline CNN by achieving approximately 9.1% lower MPJPE and 7.3% lower PA-MPJPE. This improvement suggests that self-attention mechanisms are more capable of extracting global

Table 6. Hybrid CNN-Transformer Per-Joint Standard Deviation with Modified MSE Loss

Joint	Body Part	Pred Std	GT Std
0	Pelvis (root)	0.0000	0.0000
1	Left hip	0.0498	0.0509
2	Left knee	0.2472	0.2433
3	Left ankle	0.4343	0.4303
4	Right hip	0.0591	0.0509
5	Right knee	0.1318	0.1385
6	Right ankle	0.3217	0.3268
7	Spine	0.1491	0.1426
8	Chest/Thorax	0.2868	0.2853
9	Neck	0.3373	0.3463
10	Head	0.3745	0.3733
11	Right shoulder	0.3207	0.3224
12	Right elbow	0.2860	0.3239
13	Right wrist	0.2087	0.3416
14	Left shoulder	0.1946	0.1942
15	Left elbow	0.0854	0.1603
16	Left wrist	0.0746	0.2598

structure from multi-antenna CSI signals than convolutional filters alone.

However, our results also highlight fundamental limitations in the expressiveness of the WiFi CSI modality. Both models reached a validation loss plateau early in training despite the use of learning rate scheduling and root-centered MSE loss. As noted in the MM-Fi study, WiFi-based 3D HPE models often suffer from a performance ceiling because the “significant decline in performance thanks to the limited resolution which fails to fully capture human subtle motions and differences” [8].

Our per-joint and per-action tables and analysis provide concrete evidence for this lack of resolution. We observed a failure to capture the true joint variance, with the predicted standard deviations for wrists and elbows being significantly lower than ground truth (e.g., 0.0746 vs 0.2598 for the left wrist). This poor variance representation confirms observations in Meta-Fi, where researchers found that “the prediction of arms and legs is not satisfactory, especially for the wrists, since the motions of limbs and wrists are subtle” [7]. This challenge is further reflected in our per-action results, where basic motions like raising hands (95.0 mm MPJPE) were significantly more accurate than complex motions that require depth perception like picking up things (194.8 mm MPJPE).

In summary, while the hybrid CNN-Transformer architecture advances the state-of-the-art for privacy-preserving sensing, the sparse spatial resolution of commercial WiFi remains a bottleneck for high-fidelity 3D HPE. For future work, we suggest exploring multimodal data fusion, such as combining WiFi CSI with complementary low-resolution

sensors like LiDAR or mmWave radar, to enhance limb motion capture while maintaining a privacy-first approach to human sensing.

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A. Code Implementation

We implemented our models on Google Colab using a shared drive found here: [CS 682 Final Project](#). If the permissions do not work please contact the authors.